AI Security in Action: Exploring Perplexity Bumblebee & Numbat

PROBLEM

Over the weekend, I explored R&D AI coding agents and learned about AI security tools like Numbat and Bumblebee. I learned how to scan MCP servers, configurations, and extensions for potential security risks. My goal was to understand how to make AI-assisted development more secure.

HOW I DEBUGGED IT

I installed and tested two open-source security tools: Bumblebee and Numbat on an Ubuntu EC2 instance.

- Bumblebee Scanned packages, MCP configurations, and developer extensions.
- Numbat Scanned supported AI agent artifacts for potential security issues.

THE SOLUTION

I used **Bumblebee** to check packages, MCP configurations, and developer extensions. I used **Numbat** to discover and analyze supported AI agent artifacts.

Testing both tools on Ubuntu helped me understand how to identify potential security risks in AI development environments.

EC2 → Bumblebee & Numbat → Scan → Identify Risks → Secure AI Development

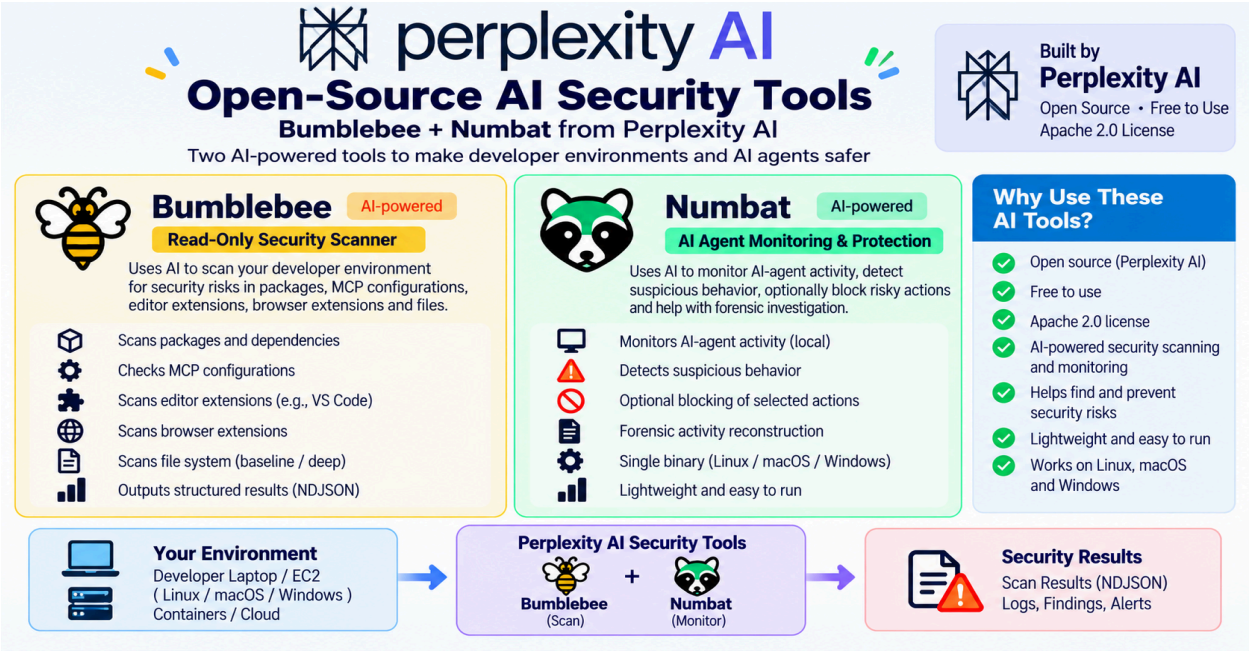
Has anyone else faced this problem?

How did you fix it? What tools do you use to secure AI coding agents?

Blog:https://vishnusai.hashnode.dev/ai-security-in-action-exploring-perplexity-bumblebee-numbat?utm_source=hashnode&utm_medium=feed

HashnodProfile:https://hashnode.com/@vishnusaiK

GitHub : https://github.com/ksaivishnusaikvs/DevOps-Engineering/tree/main/Linkedin/Post



Bumblebee is also open source and free to run. It's a read-only security scanner for developer endpoints, including packages, MCP configs, editor extensions, and browser extensions. https://github.com/perplexityai/bumblebee

Step1: Install prerequisites: **Bumblebee** requires Go 1.25+ and is a read-only scanner. GitHub

sudo apt update; sudo apt install -y git; go version

```
ubuntu@ip-172-31-29-13:~$ sudo apt update
sudo apt upgrade -y
sudo apt install -y curl wget git
Hit:1 http://us-west-2.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-west-2.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-west-2.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 https://download.docker.com/linux/ubuntu resolute InRelease
Hit:5 https://deb.nodesource.com/node_22.x nodistro InRelease
Hit:6 http://security.ubuntu.com/ubuntu resolute-security InRelease
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  docker-ce-rootless-extras  libsimdjson29              linux-headers-7.0.0-1013-aws          linux-tools-7.0.0-1013-aws  node-lru-cache
  libada-url0-3              libsimdutf31               linux-image-7.0.0-1013-aws            node-balanced-match         node-minimatch
  libcares2                  linux-aws-headers-7.0.0-1013  linux-main-modules-zfs-7.0.0-1013-aws  node-brace-expansion        node-xtend
  liblldhttp9.3              linux-aws-tools-7.0.0-1013    linux-modules-7.0.0-1013-aws          node-cjs-module-lexer       pollinate
Use 'sudo apt autoremove' to remove them.

Not upgrading yet due to phasing:
  python3-software-properties  software-properties-common

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
curl is already the newest version (8.18.0-1ubuntu2.7).
wget is already the newest version (1.25.0-2ubuntu4.4).
wget set to manually installed.
git is already the newest version (1:2.53.0-1ubuntu1).
The following packages were automatically installed and are no longer required:
  docker-ce-rootless-extras  libsimdjson29              linux-headers-7.0.0-1013-aws          linux-tools-7.0.0-1013-aws  node-lru-cache
  libada-url0-3              libsimdutf31               linux-image-7.0.0-1013-aws            node-balanced-match         node-minimatch
  libcares2                  linux-aws-headers-7.0.0-1013  linux-main-modules-zfs-7.0.0-1013-aws  node-brace-expansion        node-xtend
  liblldhttp9.3              linux-aws-tools-7.0.0-1013    linux-modules-7.0.0-1013-aws          node-cjs-module-lexer       pollinate
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
ubuntu@ip-172-31-29-13:~$ go version
Command 'go' not found, but can be installed with:
sudo apt install golang-go  # version 2:1.26-1, or
sudo apt install gccgo-go   # version 2:1.26-1
ubuntu@ip-172-31-29-13:~$ sudo apt install golang-go
The following packages were automatically installed and are no longer required:
  docker-ce-rootless-extras  libsimdjson29              linux-headers-7.0.0-1013-aws          linux-tools-7.0.0-1013-aws  node-lru-cache
  libada-url0-3              libsimdutf31               linux-image-7.0.0-1013-aws            node-balanced-match         node-minimatch
  libcares2                  linux-aws-headers-7.0.0-1013  linux-main-modules-zfs-7.0.0-1013-aws  node-brace-expansion        node-xtend
  liblldhttp9.3              linux-aws-tools-7.0.0-1013    linux-modules-7.0.0-1013-aws          node-cjs-module-lexer       pollinate
Use 'sudo apt autoremove' to remove them.

Installing:
  golang-go
```

Step 2: Install Bumblebee

go install github.com/perplexityai/bumblebee/cmd/bumblebee@latest

Add it to PATH:

echo 'export PATH="$HOME/go/bin:$PATH"' >> ~/.bashrc

source ~/.bashrc

Verify: bumblebee version

```
ubuntu@ip-172-31-29-13:~$ go install github.com/perplexityai/bumblebee/cmd/bumblebee@latest
go: downloading github.com/perplexityai/bumblebee v0.1.2
ubuntu@ip-172-31-29-13:~$ echo 'export PATH="$PATH:$HOME/go/bin"' >> ~/.bashrc
source ~/.bashrc
ubuntu@ip-172-31-29-13:~$ bumblebee version
bumblebee v0.1.2
commit: unknown
built:  unknown
go:     go1.26.0
ubuntu@ip-172-31-29-13:~$ bumblebee selftest
selftest OK (5 findings in 2ms)
ubuntu@ip-172-31-29-13:~$ bumblebee roots --profile baseline
default roots: 4 present, 123 candidate paths absent (use --root to override)
user_package_root       /home/ubuntu/go
global_package_root     /usr/local/lib
global_package_root     /usr/lib/python3
global_package_root     /usr/lib/python3.14
ubuntu@ip-172-31-29-13:~$ bumblebee scan --profile baseline > ~/bumblebee-inventory.ndjson
{"record_type":"diagnostic","record_id":"diagnostic:4db41322c28aac1bc90d1bcf20e4919244b1e13c5b85a5c5d9705e4b659973e6","run_id":"973213a546b96f0d0860f82e293b56f7","time":"20
26-10-04T16:24:41.4616917992","level":"info","message":"default roots: 4 present, 123 candidate paths absent (use --root to override)"}
{"record_type":"diagnostic","record_id":"diagnostic:e516fedc51382f971a90801066212de324d95ac600267830ce55c18eea81dba5","run_id":"973213a546b96f0d0860f82e293b56f7","time":"20
26-10-04T16:24:42.5403742392","level":"info","message":"scan complete: profile=baseline status=complete files_considered=15680 records=117 findings=0 suppressed=0 duplicate
s=0 diagnostics=1 timed_out=false duration=1.078000493s"}
ubuntu@ip-172-31-29-13:~$ bumblebee scan --profile baseline
{"record_type":"diagnostic","record_id":"diagnostic:4db41322c28aac1bc90d1bcf20e4919244b1e13c5b85a5c5d9705e4b659973e6","run_id":"954022938c07143a82cfc3df07835921","time":"20
26-10-04T16:26:31.275758685Z","level":"info","message":"default roots: 4 present, 123 candidate paths absent (use --root to override)"}
{"record_type":"package","record_id":"package:e757419d326c38a6d51b88f6d70d5e55073b705bb547543f62e54c0d2b29faa8","schema_version":"0.1.0","scanner_name":"bumblebee","scanner
_version":"v0.1.2","run_id":"954022938c07143a82cfc3df07835921","scan_time":"2026-10-04T16:26:31.275905579Z","endpoint":{"hostname":"ip-172-31-29-13","os":"linux","arch":"am
d64","username":"ubuntu","uid":"1000"},"profile":"baseline","ecosystem":"homebrew","package_name":"bumblebee-selftest-cask","normalized_name":"bumblebee-selftest-cask","ver
sion":"0.0.0","project_path":"/home/ubuntu/go/pkg/mod/github.com/perplexityai/bumblebee@v0.1.2/cmd/bumblebee/selftest/fixtures/homebrew-fixture/Caskroom","root_kind":"user_
package_root","package_manager":"homebrew","source_type":"homebrew-cask-metadata","source_file":"/home/ubuntu/go/pkg/mod/github.com/perplexityai/bumblebee@v0.1.2/cmd/bumble
bee/selftest/fixtures/homebrew-fixture/Caskroom/bumblebee-selftest-cask/.metadata/0.0.0/20260523010203.004/Casks/bumblebee-selftest-cask.json","direct_dependency":true,"has
_lifecycle_scripts":false,"confidence":"high"}
{"record_type":"package","record_id":"package:19782186921ecb7ee3a9bcd7c60951d959329cb6926f99719ea20183b470e627","schema_version":"0.1.0","scanner_name":"bumblebee","scanner
_version":"v0.1.2","run_id":"954022938c07143a82cfc3df07835921","scan_time":"2026-10-04T16:26:31.275905579Z","endpoint":{"hostname":"ip-172-31-29-13","os":"linux","arch":"am
d64","username":"ubuntu","uid":"1000"},"profile":"baseline","ecosystem":"homebrew","package_name":"bumblebee-selftest-brew","normalized_name":"bumblebee-selftest-brew","ver
sion":"0.0.0","project_path":"/home/ubuntu/go/pkg/mod/github.com/perplexityai/bumblebee@v0.1.2/cmd/bumblebee/selftest/fixtures/homebrew-fixture/Cellar","root_kind":"user_pa
ckage_root","package_manager":"homebrew","source_type":"homebrew-formula-receipt","source_file":"/home/ubuntu/go/pkg/mod/github.com/perplexityai/bumblebee@v0.1.2/cmd/bumble
bee/selftest/fixtures/homebrew-fixture/Cellar/bumblebee-selftest-brew/0.0.0/INSTALL_RECEIPT.json","direct_dependency":true,"has_lifecycle_scripts":false,"confidence":"high"
```

Step 3: Run the self-test

bumblebee selftest

Your result:

selftest OK

```
ubuntu@ip-172-31-29-13:~$ bumblebee selftest
selftest OK (5 findings in 2ms)
ubuntu@ip-172-31-29-13:~$ bumblebee roots --profile baseline
default roots: 4 present, 123 candidate paths absent (use --root to override)
```

means the installation is working.

Step 4: Baseline scan

This is the recommended lightweight inventory scan:

bumblebee scan --profile baseline > ~/bumblebee-baseline.ndjson

```
ubuntu@ip-172-31-29-13:~$ bumblebee scan \
  --profile deep \
  --root /home/ubuntu \
  > ~/bumblebee-deep.ndjson
{"record_type":"diagnostic","record_id":"diagnostic:339cf1d11077f16a0a6ba35b7fd8f50344ee1e9291ef1adc6e4ebb43e97ca35a","run_id":"af38b1e7a272addd64efeef665302f6a","time":"20
26-10-04T16:29:40.8246140722","level":"info","message":"scan complete: profile=deep status=complete files_considered=8848 records=392 findings=0 suppressed=0 duplicates=0 d
iagnostics=0 timed_out=false duration=44.934432ms"}
ubuntu@ip-172-31-29-13:~$ grep '"scan_summary"' ~/bumblebee-deep.ndjson
{"record_type":"scan_summary","record_id":"scan_summary:18b4e65e988c30389ab4fd250ed4239753ac03c5f274a1ec67b027894501166a","schema_version":"0.1.0","scanner_name":"bumblebee
","scanner_version":"v0.1.2","run_id":"af38b1e7a272addd64efeef665302f6a","scan_time":"2026-10-04T16:29:40.779432288Z","end_time":"2026-10-04T16:29:40.824530682Z","endpoint"
:{"hostname":"ip-172-31-29-13","os":"linux","arch":"amd64","username":"ubuntu","uid":"1000"},"profile":"deep","status":"complete","roots":[{"path":"/home/ubuntu","kind":"de
ep_home_root"}],"counts":{"finding":0,"package":392},"package_records_emitted":392,"findings_emitted":0,"duplicates":0,"diagnostics_count":0,"files_considered":8848,"timed_
out":false,"duration_ms":44}
ubuntu@ip-172-31-29-13:~$ sudo -u ubuntu bumblebee scan \
```

Check the final summary:

grep '"record_type":"scan_summary"' ~/bumblebee-baseline.ndjson

Bumblebee's output is NDJSON. Diagnostics are written separately to stderr. GitHub

Step5. Deep scan

For an on-demand scan of specific directories:

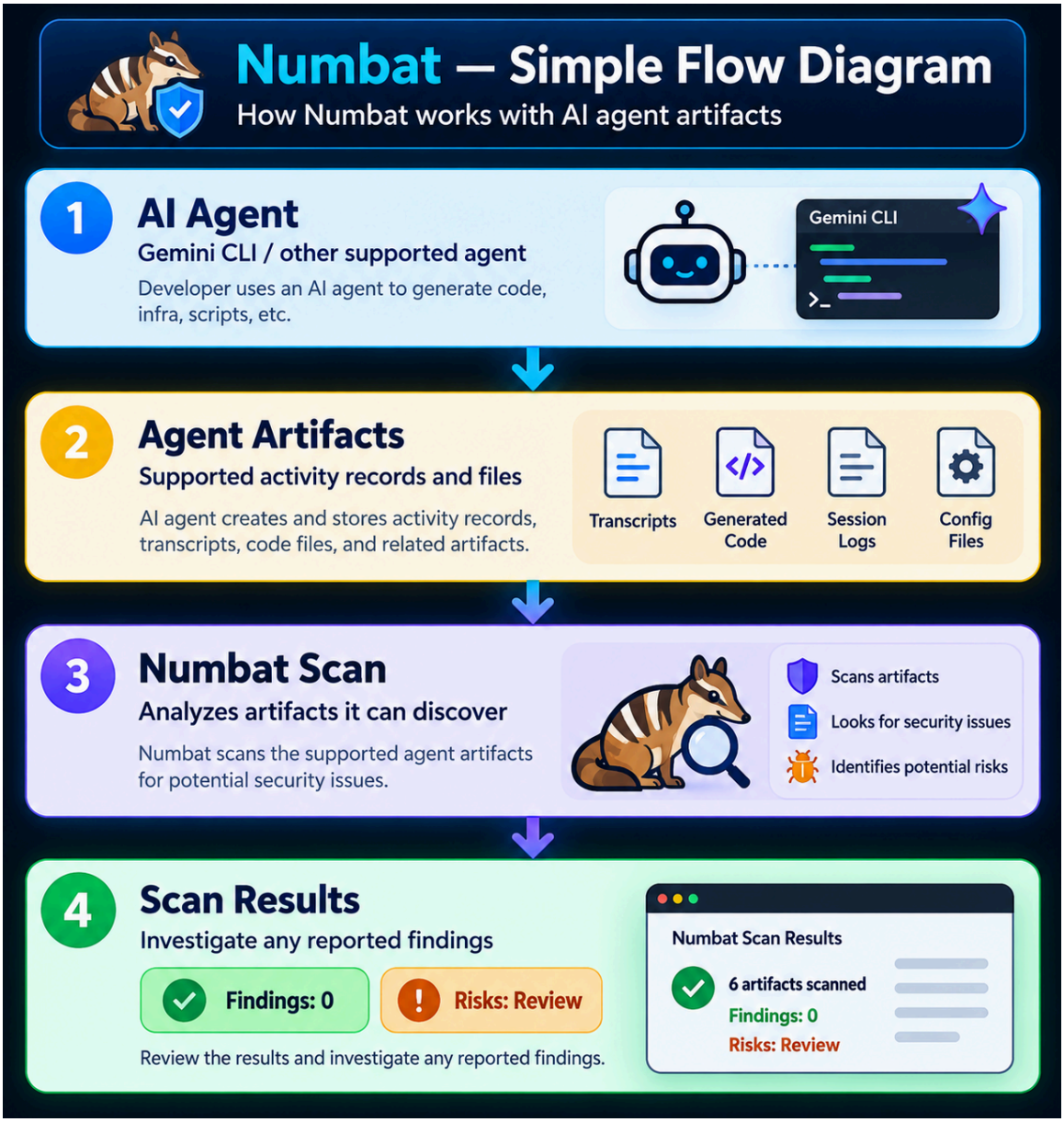
```
bumblebee scan \
  --profile deep \
  --root /home/ubuntu \
  --root /opt \
  --root /srv \
  > ~/bumblebee-deep.ndjson
```



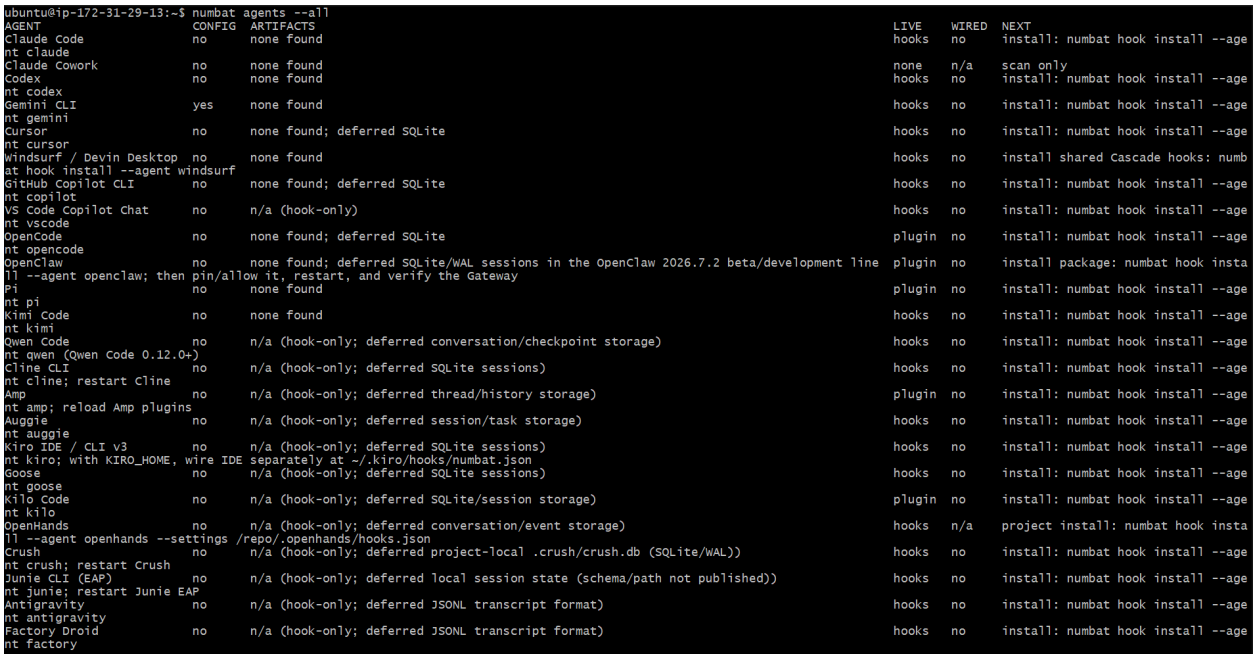
deep is specifically designed for explicit --root paths.

Numbat https://github.com/perplexityai/numbat

is the more interesting one for a DevOps/security project: it's a single binary for Linux, macOS, and Windows, and provides AI-agent monitoring, local detection, optional blocking, and forensic reconstruction. Its GitHub repo is Apache 2.0.



Step 1: numbat agents --all. Select one agent based on the scan



```
ubuntu@ip-172-31-29-13:~$ numbat agents --all
AGENT                   CONFIG  ARTIFACTS                                                                    LIVE    WIRED  NEXT
Claude Code             no      none found                                                                   hooks   no     install: numbat hook install --age
nt claude
Claude Cowork           no      none found                                                                   none    n/a    scan only
Codex                   no      none found                                                                   hooks   no     install: numbat hook install --age
nt codex
Gemini CLI              yes     none found                                                                   hooks   no     install: numbat hook install --age
nt gemini
Cursor                  no      none found; deferred SQLite                                                  hooks   no     install: numbat hook install --age
nt cursor
Windsurf / Devin Desktop  no    none found                                                                   hooks   no     install shared Cascade hooks: numb
at hook install --agent windsurf
GitHub Copilot CLI      no      none found; deferred SQLite                                                  hooks   no     install: numbat hook install --age
nt copilot
VS Code Copilot Chat    no      n/a (hook-only)                                                              hooks   no     install: numbat hook install --age
nt vscode
OpenCode                no      none found; deferred SQLite                                                  plugin  no     install: numbat hook install --age
nt opencode
OpenClaw                no      none found; deferred SQLite/WAL sessions in the OpenClaw 2026.7.2 beta/development line  plugin  no     install package: numbat hook insta
ll --agent openclaw; then pin/allow it, restart, and verify the Gateway
Pi                      no      none found                                                                   plugin  no     install: numbat hook install --age
nt pi
Kimi Code               no      none found                                                                   hooks   no     install: numbat hook install --age
nt kimi
Qwen Code               no      n/a (hook-only; deferred conversation/checkpoint storage)                    hooks   no     install: numbat hook install --age
nt qwen (Qwen Code 0.12.0+)
Cline CLI               no      n/a (hook-only; deferred SQLite sessions)                                    hooks   no     install: numbat hook install --age
nt cline; restart Cline
Amp                     no      n/a (hook-only; deferred thread/history storage)                             plugin  no     install: numbat hook install --age
nt amp; reload Amp plugins
Auggie                  no      n/a (hook-only; deferred session/task storage)                               hooks   no     install: numbat hook install --age
nt auggie
Kiro IDE / CLI v3       no      n/a (hook-only; deferred SQLite sessions)                                    hooks   no     install: numbat hook install --age
nt kiro; with KIRO_HOME, wire IDE separately at ~/.kiro/hooks/numbat.json
Goose                   no      n/a (hook-only; deferred SQLite sessions)                                    hooks   no     install: numbat hook install --age
nt goose
Kilo Code               no      n/a (hook-only; deferred SQLite/session storage)                             plugin  no     install: numbat hook install --age
nt kilo
OpenHands               no      n/a (hook-only; deferred conversation/event storage)                         hooks   n/a    project install: numbat hook insta
ll --agent openhands --settings /repo/.openhands/hooks.json
Crush                   no      n/a (hook-only; deferred project-local .crush/crush.db (SQLite/WAL))         hooks   no     install: numbat hook install --age
nt crush; restart Crush
Junie CLI (EAP)         no      n/a (hook-only; deferred local session state (schema/path not published))    hooks   no     install: numbat hook install --age
nt junie; restart Junie EAP
Antigravity             no      n/a (hook-only; deferred JSONL transcript format)                            hooks   no     install: numbat hook install --age
nt antigravity
Factory Droid           no      n/a (hook-only; deferred JSONL transcript format)                            hooks   no     install: numbat hook install --age
nt factory
```

Step2: Login to the Gemini CLI command gemini shown below



```
nt hermes
ubuntu@ip-172-31-29-13:~$ numbat hook install --agent gemini --emit all
gemini      ok          installed numbat hooks (/home/ubuntu/.gemini/settings.json)
ubuntu@ip-172-31-29-13:~$ numbat hook status --agent gemini
gemini      installed   numbat hooks installed (/home/ubuntu/.gemini/settings.json)
ubuntu@ip-172-31-29-13:~$ numbat hook status --agent gemini
gemini      installed   numbat hooks installed (/home/ubuntu/.gemini/settings.json)
ubuntu@ip-172-31-29-13:~$ numbat agents
AGENT       CONFIG  ARTIFACTS   LIVE    WIRED  NEXT
Gemini CLI  yes     none found  hooks   yes    install: numbat hook install --agent gemini
ubuntu@ip-172-31-29-13:~$ gemini

 ▝▜▄  GEMINI

 Gemini CLI v0.62.0


Tips for getting started:
1. Create GEMINI.md files to customize your interactions
2. /help for more information
3. Ask coding questions, edit code or run commands
4. Be specific for the best results

 Skipping project agents due to untrusted folder. To enable, ensure that the project root is trusted.

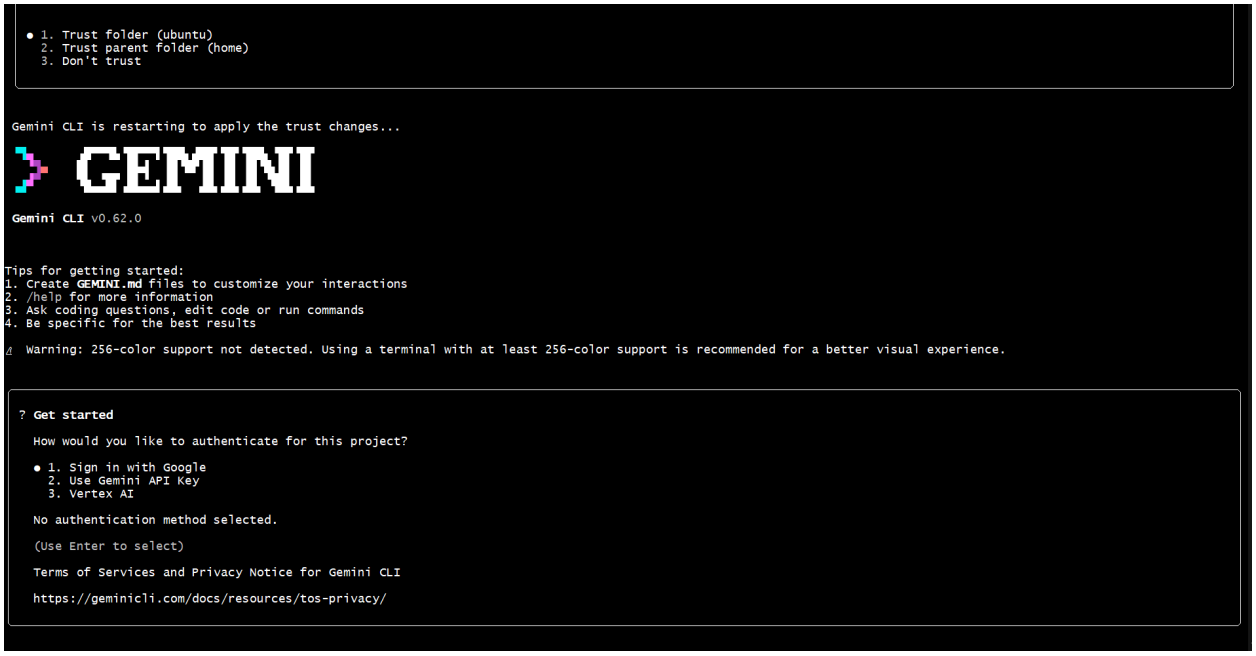
⚠  Warning you are running Gemini CLI in your home directory.
   This warning can be disabled in /settings
⚠  Warning: 256-color support not detected. Using a terminal with at least 256-color support is recommended for a better visual experience.


  Do you trust the files in this folder?

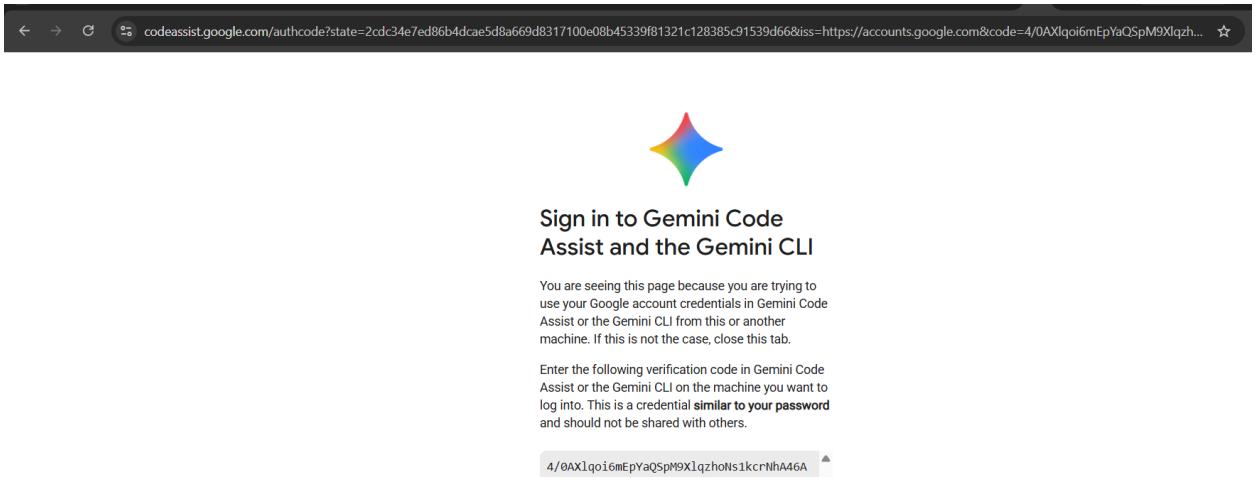
  Trusting a folder allows Gemini CLI to load its local configurations, including custom commands, hooks, MCP servers, agent skills, and settings. These configurations
  could execute code on your behalf or change the behavior of the CLI.


  ● 1. Trust folder (ubuntu)
    2. Trust parent folder (home)
    3. Don't trust
```

Step3 : Click on the Sign in with Google area; select based on your options. I selected 1 option: Sign in with Google



Click Sign in it shows the URL in the web, then we can log in



Step3 : Test Numbat in the first CLI; it does not show **Image 1:** no agent artifacts to analyze in that dir. **Image 2:** Numbat Agent Scan: Numbat successfully discovered and parsed 6 artifacts with 0 findings.

Image 1: Gemini CLI Integration: Numbat scanned the DevOps directory but found no agent artifacts to analyze. The scan returned an error because it found no supported artifacts.

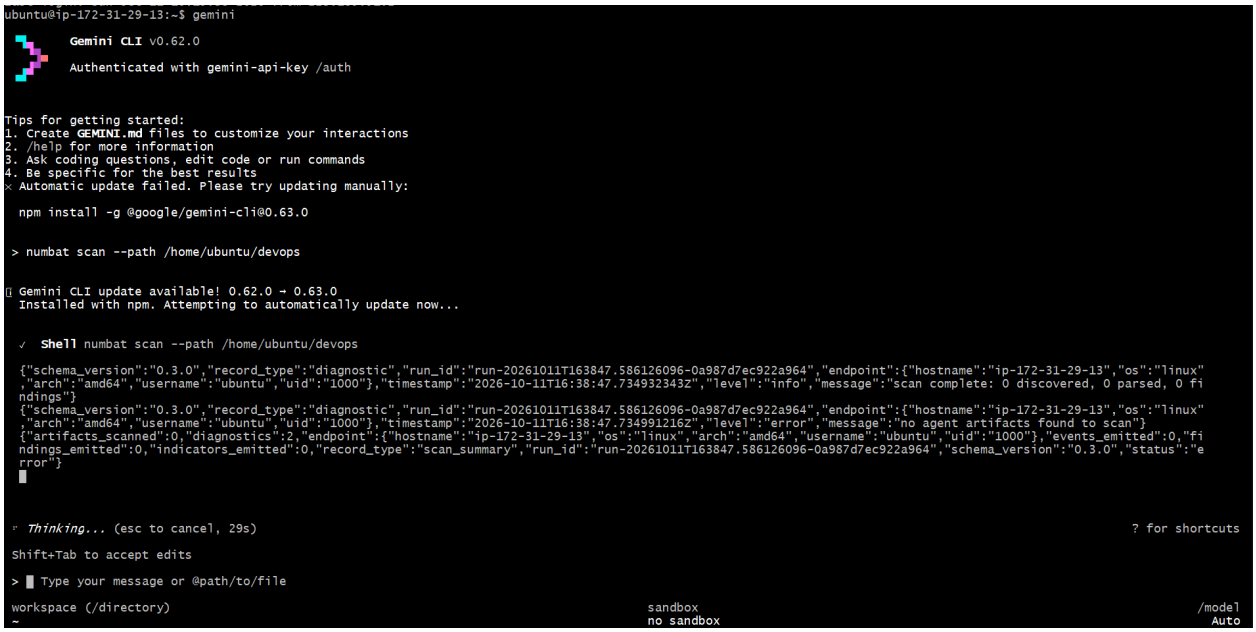
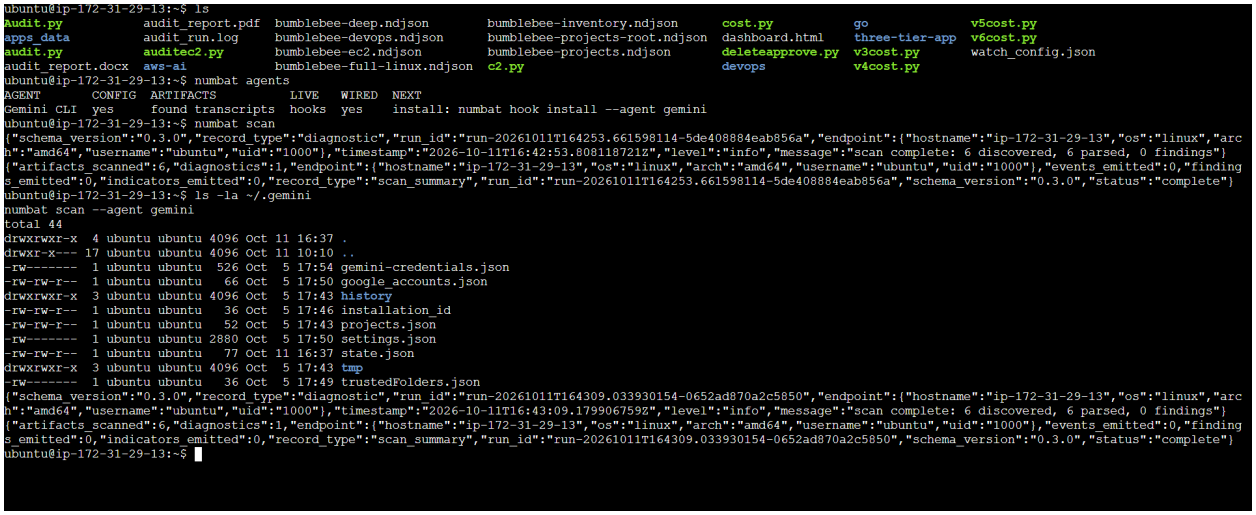


Image 2: Numbat Agent Scan: Numbat successfully discovered and parsed 6 artifacts with 0 findings. The scan completed successfully, confirming that artifact discovery and parsing worked in this run.



```
ubuntu@ip-172-31-29-13:~$ numbat agents --all
AGENT                  CONFIG  ARTIFACTS                                                
```